

# SAFETY DATA SHEET

Revision date 24-May-2023

Revision Number 1.1

## Section 1: Identification

### Product identifier

**Product Name** 2% Bis Solution  
**Catalogue Number(s)** 1610142, 1610143, 1610142EDU

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Albany Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

### GHS Classification

|                                                         |             |
|---------------------------------------------------------|-------------|
| <b>Germ cell mutagenicity</b>                           | Category 1B |
| <b>Carcinogenicity</b>                                  | Category 1B |
| <b>Reproductive toxicity</b>                            | Category 1B |
| <b>Specific target organ toxicity — single exposure</b> | Category 1  |

### Label elements



#### **Signal word**

Danger

#### **Hazard statements**

May cause genetic defects  
May cause cancer  
May damage fertility or the unborn child  
Causes damage to organs

#### **Precautionary Statements - Prevention**

Do not handle until all safety precautions have been read and understood  
Wear protective gloves/clothing and eye/face protection  
Do not breathe dust/fume/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

**Precautionary Statements - Response**

IF exposed or concerned: Call a POISON CENTER or doctor

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

May be harmful if swallowed.

**Section 3: Composition/information on ingredients**

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Water                     | 7732-18-5   | 50 - 100 |
| Methylene diacrylamide    | 110-26-9    | 1 - 2.5  |
| Non-hazardous ingredients | Proprietary | Balance  |

**Section 4: First-aid measures****Description of first aid measures**

|                       |                                                                                                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b> | Show this safety data sheet to the doctor in attendance. IF exposed or concerned: Get medical advice/attention.                                                         |
| <b>Inhalation</b>     | Remove to fresh air. IF exposed or concerned: Get medical advice/attention.                                                                                             |
| <b>Eye contact</b>    | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If symptoms persist, call a doctor.                                            |
| <b>Skin contact</b>   | Wash off immediately with plenty of water for at least 15 minutes. If symptoms persist, call a doctor.                                                                  |
| <b>Ingestion</b>      | Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Call a doctor. Clean mouth with water and drink afterwards plenty of water. |

**Most important symptoms and effects, both acute and delayed**

|                 |                           |
|-----------------|---------------------------|
| <b>Symptoms</b> | No information available. |
|-----------------|---------------------------|

**Indication of any immediate medical attention and special treatment needed**

|                        |                        |
|------------------------|------------------------|
| <b>Note to doctors</b> | Treat symptomatically. |
|------------------------|------------------------|

**Section 5: Fire-fighting measures****Suitable Extinguishing Media**

|                                     |                                                                                                         |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b> | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| <b>Large Fire</b>                   | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical** No information available.

**Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.

## Section 6: Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Remove contaminated clothing and shoes.

**General hygiene considerations** Do not eat, drink or smoke when using this product. Wash hands before breaks and immediately after handling the product.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store locked up. Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## Section 8: Exposure controls/personal protection

**Control parameters**

|                                                                                     |                                                                                                                                                                          |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b>                                                              | This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.              |
| <b>Biological occupational exposure limits</b>                                      | This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.                         |
| <b><u>Appropriate engineering controls</u></b>                                      |                                                                                                                                                                          |
| <b>Engineering controls</b>                                                         | Showers<br>Eyewash stations<br>Ventilation systems.                                                                                                                      |
| <b><u>Individual protection measures, such as personal protective equipment</u></b> |                                                                                                                                                                          |
| <b>Eye/face protection</b>                                                          | No special protective equipment required.                                                                                                                                |
| <b>Hand protection</b>                                                              | Wear suitable gloves.                                                                                                                                                    |
| <b>Skin and body protection</b>                                                     | Wear suitable protective clothing.                                                                                                                                       |
| <b>Respiratory protection</b>                                                       | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>Environmental exposure controls</b>                                              | No information available.                                                                                                                                                |

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <b><u>Property</u></b>                        | <b><u>Values</u></b>      | <b><u>Remarks • Method</u></b> |
|-----------------------------------------------|---------------------------|--------------------------------|
| <b>pH</b>                                     |                           | None known                     |
| <b>Melting point / freezing point</b>         | No data available         | None known                     |
| <b>Boiling point / boiling range</b>          | No data available         | None known                     |
| <b>Flash point</b>                            | No data available         | None known                     |
| <b>Evaporation rate</b>                       | No data available         | None known                     |
| <b>Flammability (solid, gas)</b>              | No data available         | None known                     |
| <b>Flammability Limit in Air</b>              |                           | None known                     |
| <b>Upper flammability or explosive limits</b> | No data available         |                                |
| <b>Lower flammability or explosive limits</b> | No data available         |                                |
| <b>Vapour pressure</b>                        | No data available         | None known                     |
| <b>Vapour density</b>                         | No data available         | None known                     |
| <b>Relative density</b>                       | 1.000                     | None known                     |
| <b>Water solubility</b>                       | No data available         | Miscible in water              |
| <b>Solubility(ies)</b>                        | No data available         | None known                     |
| <b>Partition coefficient</b>                  | No data available         | None known                     |
| <b>Autoignition temperature</b>               | No data available         | None known                     |
| <b>Decomposition temperature</b>              |                           | None known                     |
| <b>Kinematic viscosity</b>                    | No data available         | None known                     |
| <b>Dynamic viscosity</b>                      | No data available         | None known                     |
| <b>Explosive properties</b>                   | No information available. |                                |
| <b>Oxidising properties</b>                   | No information available. |                                |

### Other information

|                          |                          |
|--------------------------|--------------------------|
| Softening point          | No information available |
| Molecular weight         | No information available |
| VOC content              | No information available |
| Liquid Density           | No information available |
| Bulk density             | No information available |
| Particle characteristics | No information available |

## Section 10: Stability and reactivity

### Reactivity

**Reactivity** No information available.

### Chemical stability

**Stability** Stable under normal conditions.

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

### Incompatible materials

**Incompatible materials** None known based on information supplied.

### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

#### Product Information

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** May be harmful if swallowed.

**Symptoms** No information available.

### Acute toxicity

#### Numerical measures of toxicity

The following values are calculated based on chapter 3.1 of the GHS document

|                 |                 |
|-----------------|-----------------|
| ATEmix (oral)   | 5,000.00 mg/kg  |
| ATEmix (dermal) | 57,050.00 mg/kg |

**Component Information**

| Chemical name          | Oral LD50           | Dermal LD50 | Inhalation LC50 |
|------------------------|---------------------|-------------|-----------------|
| Water                  | > 90 mL/kg ( Rat )  | -           | -               |
| Methylene diacrylamide | = 390 mg/kg ( Rat ) | -           | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                                 |                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>                | No information available.                                                                                                                                                                                                                                                                |
| <b>Serious eye damage/eye irritation</b>        | No information available.                                                                                                                                                                                                                                                                |
| <b>Respiratory or skin sensitisation</b>        | No information available.                                                                                                                                                                                                                                                                |
| <b>Germ cell mutagenicity</b>                   | Contains a known or suspected mutagen. Classification based on data available for ingredients. May cause genetic defects.                                                                                                                                                                |
| <b>Carcinogenicity</b>                          | Contains a known or suspected carcinogen. Classification based on data available for ingredients. May cause cancer.                                                                                                                                                                      |
| <b>Reproductive toxicity</b>                    | Contains a known or suspected reproductive toxin. Classification based on data available for ingredients. May damage fertility or the unborn child.                                                                                                                                      |
| <b>STOT - single exposure</b>                   | Based on the classification criteria of the Globally Harmonized System as adopted in the country or region with which this safety data sheet complies, this product has been determined to cause systemic target organ toxicity from acute exposure. (STOT SE). Causes damage to organs. |
| <b>STOT - repeated exposure</b>                 | No information available.                                                                                                                                                                                                                                                                |
| <b>Aspiration hazard</b>                        | No information available.                                                                                                                                                                                                                                                                |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS.                                                                                                                                                                                          |

**Section 12: Ecological information****Ecotoxicity**

|                                 |                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>      | The environmental impact of this product has not been fully investigated.                  |
| <b>Unknown aquatic toxicity</b> | 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |
| <b>Terrestrial ecotoxicity</b>  | There is no data for this product.                                                         |

**Persistence and degradability** No information available.

### Bioaccumulative potential

#### Bioaccumulation

##### Component Information

| Chemical name          | Partition coefficient |
|------------------------|-----------------------|
| Methylene diacrylamide | -0.08                 |

### Mobility in soil

**Mobility** No information available.

### Other adverse effects

No information available.

## Section 13: Disposal considerations

### Waste treatment methods

#### Waste from residues/unused products

Dispose of product in packaging in a way that is consistent with the EPA Consolidation 30 April 2021 of the Hazardous Substances (Disposal) Notice 2017 and the Act.  
Treat the substance using a method that changes the characteristics or composition of the substance so that the substance is no longer a hazardous substance; or export the substance from New Zealand as waste.  
Substances which are hazardous to human health or corrosive to metals – may be discharged into the environment if a tolerable exposure limit has been set for the substance (or a component of that substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the tolerable exposure limit. If there is no tolerable exposure limit for the substance, then it may only be discharged into the environment if the substance is very rapidly converted to substances that are not hazardous substances.  
Dispose of in accordance with local regulations.  
Dispose of waste in accordance with environmental legislation.

#### Contaminated packaging

For packages that have been in direct contact with hazardous substances, the person must ensure that the package is rendered incapable of containing any substance. It must be disposed of in a manner that is consistent with the requirements for disposal of the substance that it contained, taking into account the material the package is manufactured from.  
Packages may only be reused or recycled if:  
- the substance has a physical hazard other than corrosive to metal, and has been treated to remove any residual contents of the hazardous substance;  
- or for substances that have a health or environmental hazard, or corrosive to metal, the contents of the residue in the package are below the threshold for the substance to be classified as hazardous in the Hazardous Substances (Hazard Classification) Notice 2020.

## Section 14: Transport information

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

#### Special precautions for user

Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

**EPA New Zealand HSNO approval code or group standard** To be determined

#### **National regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDSL</b>      | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AICS</b>          | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances

**IECSC** - China Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

## Section 16: Other information

#### **Revision date**

24-May-2023

#### **Revision Note**

Reformatted and updated existing information.

#### **Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION



---

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**